Sec: JEE APEX & SR IIT MODEL-A | DT: 31/12/2025
Time: 3 Hrs | GTM – 06 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
B	**A**	**B**	**A**	**C**	**A**	**A**	**A**	**C**	**D**
11	12	13	14	15	16	17	18	19	20
A	**D**	**D**	**A**	**C**	**A**	**C**	**A**	**C**	**B**
21	22	23	24	25					
5	**16**	**1**	**11**	**312**					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	**A**	**C**	**D**	**D**	D	**C**	**B**	**D**	**C**
36	37	38	39	40	41	42	43	44	45
C	**D**	**C**	**C**	**B**	B	**B**	**D**	**A**	**C**
46	47	48	49	50					
1	**16**	**2**	**123**	**392**					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	**A**	**B**	**C**	**B**	A	**A**	**A**	**B**	**D**
61	62	63	64	65	66	67	68	69	70
B	D	**C**	**D**	**B**	**C**	C	**A**	**D**	**B**
71	72	73	74	75					
6	**3**	**3**	**5**	5					

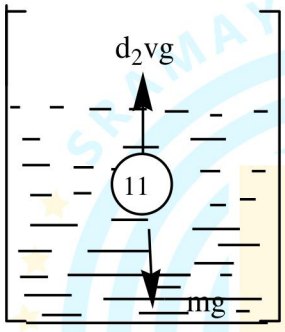
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HINTS
PHYSICS

1. Potential drops across C_1 and C_2 are 6V and 4V, respectively. Since they are in series, same charge flows through them and

$$\frac{C_1}{C_2}=\frac{V_2}{V_1}=\frac{2}{3}$$

2.



$F :\rightarrow$ viscous force

$F+d_2v_g=mg$

$F=mg-d_2v_g$

$$=mg\left[1-\frac{d_2v_g}{mg}\right]$$

$$=mg\left[1-\frac{d_2v_g}{d_1\cancel{v}_{\cancel{g}}}\right]$$

$$=mg\left[1-\frac{d_2}{d_1}\right]$$

$$V_T=\frac{2}{9}r^2\frac{(d_2-d_1)g}{\eta}$$

3. Conceptual

4. Since $\vec{E}=-\Delta V$

$$\frac{dV}{dx}=-yz\rightarrow(i)$$

$$\frac{dV}{dz}=xy\rightarrow(ii)$$

After integrating Eq (i)

$$V=-xyz+f(y,z)$$

Differentiating w.r.t z:

$$\frac{dV}{dZ}=-xy+\frac{d(y)}{dz}$$

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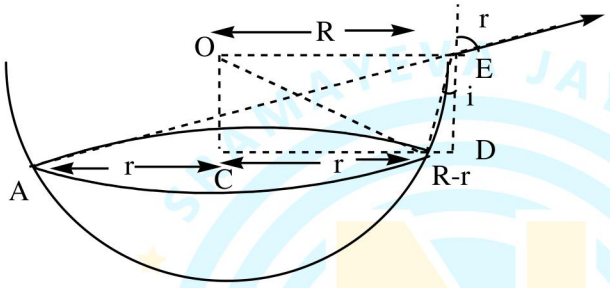
After comparing: $\frac{dg}{dz}=0$

So, $V=-xyz$

So, equations of equipotential surface

$xyz=$ constant

5.



DE=OC

$=\sqrt{(OB)^2-(BC)^2}$

$=\sqrt{R^2-r^2}$

$\sin i=\frac{BD}{BE}=\frac{R-r}{\sqrt{(R-r)^2+\left(\sqrt{R^2-r^2}\right)^2}}$

$\sin i=\frac{R-r}{\sqrt{R^2-r^2-2Rr+R^2-r^2}}$

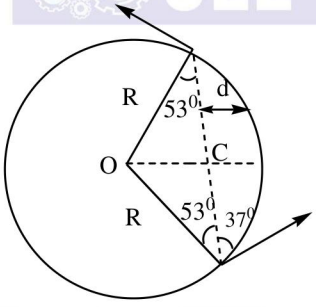
$=\frac{R-r}{\sqrt{2R^2-2Rr}}$

$=\frac{R-r}{\sqrt{2R}\left(\sqrt{R-r}\right)}=\sqrt{\frac{R-r}{2R}}$

$\sin r=\frac{AD}{AC}$ similarly $\sin r=\sqrt{\frac{R+r}{2R}}$

So, $\mu=\frac{\sin i}{\sin r}$ after solving: $r=R\frac{\left(\mu^2-1\right)}{\left(\mu^2+1\right)}$

6.



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$$OC = R\sin 53^0$$

$$= R \times \frac{4}{5} = \frac{4r}{5}$$

$$d = R - \frac{4r}{5} = \frac{R}{5}$$

$$= \frac{mv}{QB \times 5}$$

$$= \frac{10^{-6} \times \cancel{50}^{\cancel{10}}}{10^{-6} \times \cancel{10} \times \cancel{5}}$$

$$= 1m$$

7.

As, $\Delta W_{AB} = p\Delta V = 10(2-1) = 10\ \text{J}$

and $\Delta W_{BC} = 0$, because V is constant

From first law of thermodynamics,

$$\Delta Q = \Delta W + \Delta U$$

As $ABCA$ is a cyclic process, therefore

$$\Delta U = 0$$

$$\therefore \quad \Delta Q = \Delta W_{AB} + \Delta W_{BC} + \Delta W_{CA}$$

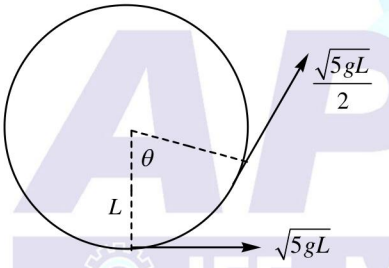
$$= \Delta W_{AB} + \Delta W_{CA}$$

or $$\Delta W_{CA} = \Delta Q - \Delta W_{AB} = 5 - 10 = -5\ \text{J}$$

8. Since velocity of sound is directly proportional to $\sqrt{T}$ So, as temperature increases velocity increases.

9. $V_{RMS} = \sqrt{\frac{3RT}{M}}$; $V^2{}_{RMS} = \frac{3RT}{M}$ So, $V^2{}_{RMS} \alpha \frac{1}{M}$

10.



$$\frac{1}{2}mv^2 = \frac{1}{2}m \times \left(\frac{v}{2}\right)^2 + mgh$$

$$\frac{1}{2}m \times 5gl = \frac{1}{2}m \times \frac{5gL}{4} + mgh$$

$$\frac{1}{2}\cancel{m} \times 5gL \times \frac{3}{4} = \cancel{m}\ \cancel{g}\ \cancel{L}\left(1 - \cos\theta\right)$$

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$$\cos\theta = \frac{-7}{8} \text{ or } \theta = \cos^{-1}\left(\frac{7}{8}\right)$$

So, $\frac{3\pi}{4} < \theta < \pi$

11. Friction is a non-conservative force so its work done will go into head

12. $\frac{A}{Z} \times \xrightarrow{4\alpha, 2B^-} \frac{206}{82}\text{pb} + 4\frac{4}{2}\text{He} + 2\frac{0}{-1}\text{B}$

$A = 206 + 16 = 222$

$Z = 82 + 8 - 2 = 88$

13. If $A = 1, B = 0$

Output through and gate=0

And if C=1;

Output through OR gate =1

14. Resistance of each part = R/2n. For n such parts connected in series, equivalent resistances, say

$$R_1 = n = \left[\frac{R}{2n}\right] = \frac{R}{2}$$

Similarly, equivalent resistance, say R_2, for another set of n identical, respectively, in parallel resistances would be

$$\frac{1}{n}\left(\frac{R}{2n}\right) = \frac{R}{2n^2}$$

For getting maximum of R_1 and R_2, the resistances should be connected in series and hence,

$$R_{eq} = R_1 + R_2 = \frac{R}{2}\left(1 + \frac{1}{n^2}\right)$$

15. There is symmetry in the square base between B and C

For equivalent resistance between A and D

$AB = BC = CD = DA = r$

$OA = OB = OC = OD = r$

About A and B we can consider Band C as common point (E)

If we combine B and C

Then AB and A to C will be in parallel

$$R_{AE} = \frac{r}{2},\ R_{OE} = \frac{r}{2},\ R_{DE} = r$$

$$R_{OA} = r;\ R_{AD} = \frac{8r}{15}$$

And similarly $R_{AO} = \frac{7r}{15}$

16. Initial energy density $= \frac{1}{2}\epsilon_0 (36)$

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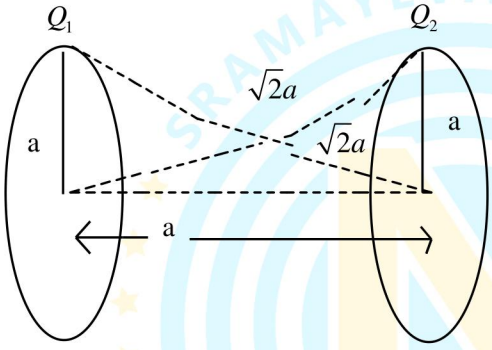
Final energy density $= 2 \times \frac{1}{2} \in_0 (36)^2$

Let`s take final electric field $= E'$

Then: $\frac{1}{2} \in_0 E'^2 = 2 \times \frac{1}{2} \in_0 (36)^2$

$E = 6\sqrt{2}$

17.



$$V_1 = \frac{1}{4\pi \in_0} \frac{Q_1}{a} + \frac{1}{4\pi \in_0} \frac{Q_2}{a\sqrt{2}}$$

Similarly for 2nd ring (at centre)

$$V_2 = \frac{1}{4\pi \in_0} \frac{Q_1}{a\sqrt{2}} + \frac{1}{4\pi \in_0} \frac{Q_2}{a}$$

Work done $= Q(V_1 - V_2)$

$$= Q \frac{(\sqrt{2}-1)}{4\pi \in_0 a\sqrt{2}} (Q_1 - Q_2)$$

18. Kinetic energy gained by a charge a after acceleration through a potential difference V_0 volt

$$QV_0 = \frac{1}{2} mV^2$$

$$V = \sqrt{\frac{2QV_0}{m}}$$

So, $mv = \sqrt{2mQV_0}$

$$\lambda = \frac{h}{mV} = \frac{h}{\sqrt{2mQV_0}}$$

$$\frac{\lambda p}{\lambda \alpha} = \sqrt{\frac{4 \times 2}{1 \times 1}} = 2\sqrt{2}$$

19. $\mu = a + \frac{bx}{L} \quad d\mu = \left(a + \frac{bx}{L}\right) dx$

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$$\times_{com} = \frac{\int dm.x}{\int dm}$$

$$\frac{7L}{12} = \frac{\int_0^L \left(a + \frac{bx}{L}\right) x\,dx}{\int_0^L \left(a + \frac{bx}{L}\right) dx}$$ after/ calculation/ a=2b

20.

Following arguments lead us easily to the right choice.

(i) Angle between any two lines is the same as the angle between their perpendiculars.

$\therefore \quad i = 30°$

(ii) $\frac{1}{1.5} = \frac{\sin 30°}{\sin r}$ (by Snell's law)

or $\sin r = \frac{1.5}{2} = 0.75$

or $r = 48.6°$

(iii) $\theta = r - i = 18.6°$

$\therefore$ Required angle $= 2 \times 18.6 = 37.2°$

21. pitch=1mm

No. of circular scale=200

So, Least count $= \frac{1mm}{200} = 0.005mm$

Initial reading:-

MSR=2division

CSR=28 division

Initial position= MSR×P+CSR×LC

$= 2mm + 28 \times 0.005 = 2.14mm$

After compression:

Angle turned $= 180^0$

Division turned $= \frac{18^0}{360^0} \times 200 = 10$ division

So, compression (x) $= 10 \times 0.005mm$

$= 0.05mm$

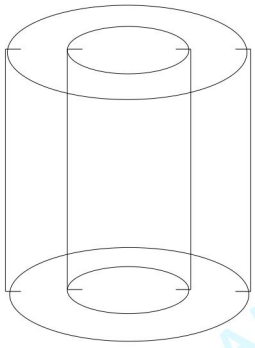
$= 0.00005m$

So, force extended $= 100 \times 0.00005$

$= 0.005N = 5$ mN

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22.



$$I = \frac{M_1 \times \left(\frac{20}{100}\right)^2}{2} - \frac{M_2 \times \left(\frac{10}{100}\right)^2}{2}$$

Here $M_1 = \frac{M}{\pi \times \left(\left(\frac{20}{100}\right)^2 - \left(\frac{10}{100}\right)^2\right) h} \times \pi \times \left(\frac{20}{100}\right)^2 \times h$

$$M_2 = \frac{M \times \pi \times \left(\frac{10}{100}\right)^2 \times h}{\pi \times \left[\left(\frac{20}{200}\right)^2 - \left(\frac{10}{100}\right)^2\right] \times h}$$

Now $MR^{'2} = I$ calculate R'

23. Dimensional formula for angular momentum

Per unit mass $= M^0 l^2 T^{-1}$

So, a=0, b=2, c=-1

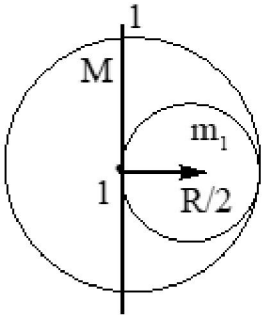
$a + b + c = 0 + 2 - 1$

=1

24. Mass of cut disc: $m_1 = M/4$

Moment of inertia of original disc about axis 1:

$$I = \frac{1}{4} MR^2$$



Moment of inertia of small disc about axis 1:

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$$I' = \frac{1}{4}m_1\left(\frac{R}{2}\right)^2 + m_1\left(\frac{R}{2}\right)^2$$

$$= \frac{5}{16}m_1R^2 = \frac{5}{64}MR^2$$

Required moment of inertia:

$$I - I' = \frac{1}{4}MR^2 - \frac{5}{64}MR^2 = \frac{11}{64}MR^2$$

25. d=2mm $= 2\times10^{-3}\,m$

$D = 1.2m$

$\lambda_1 = 6500\times10^{-10} = 6.5\times10^{-7}\,m$

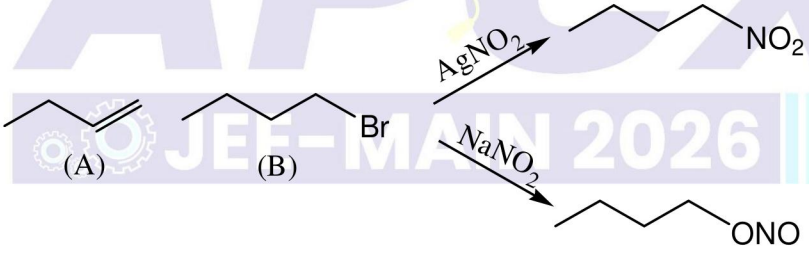
$\lambda_2 = 5200\times10^{-10} = 5.2\times10^{-7}\,m$

Distance of xth bright fringe

$$x_n = \frac{n\lambda D}{d} \qquad x_d = \frac{4\lambda_1 D}{d}\Big| x_d^{'} = \frac{4\lambda_c D}{d}$$

$D_x = x_d - x_d^{'} = 0.312mm$

CHEMISTRY

26. Conceptual
27. Conceptual
28. All carboxylic acids and dinitro, polynitro phenol can liberate Co_2g with $NaHCO_3$
29. Conceptual
30. Conceptual
31. Conceptual
32. Ketones can not give Fehling's solution, $\alpha - CH_3$ group of ketones can give iodoform
33. (b) In second orbit no d-orbital possible
    (c) for $\ell = 0, m = 0$
34. 

35. $\Delta T_f = K_f.m$

$\Delta T_b = K_b.m$

$\Delta T_b + \Delta T_f = m(K_b + K_f)$

$BP_{solution} - BP_{solvent} + FP_{solvent} - FP_{solution} = m(K_f + K_b)$

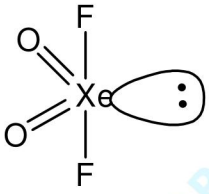
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$BP_{solution} - 100 + 0 - FP_{solution} = m(1.86 + 0.52)$

$104.76 - 100 = \frac{w}{60}\frac{1000}{100}(2.38)$

$W = 12gms$

36.



$4 + 1.LP = 5sp^3d$

37. $r = k[X]^1[Y]^0$

i) $2 \times 10^{-3} = k(0.1)$

$\therefore k = 2 \times 10^{-2}$

$t_{1/2} = \frac{0.693}{k} = 34.65\,min$

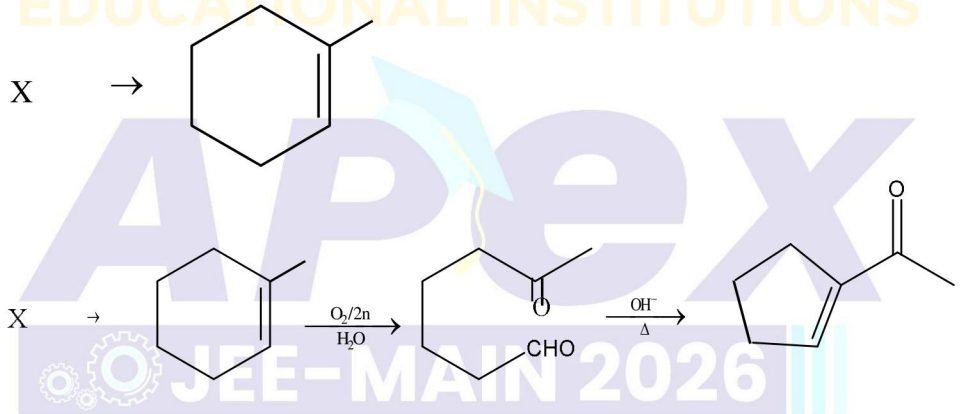
ii) $r = 4 \times 10^{-3} = [L]\left(2 \times 10^{-2}\right)$

$[L] = 0.2$

iii) $M \times 10^{-3} = \left(2 \times 10^{-2}\right)(0.4) = 8 \times 10^{-3}$

$\frac{M}{L} = \frac{8}{0.2} = 40$

38.



40. SO_2 – Bent

SF_4 – See – Saw

ClF_3 – T – Shape

BrF_5 – Square pyramidal

XeF_4 – Square planar

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41. Decreasing order of ionic radii is $C^{-4} > N^{-3} > O^{-2}$

42. Conceptual

43. Conceptual

44. Conceptual

45. $pH = 4 \qquad \left[H^+\right] = 10^{-4}$

$\left[H^+\right] = \sqrt{k_a.c}$

$\left(10^{-4}\right)^2 = ka.10^{-1}$

$ka = 10^{-7}$

$N_1V_1 = N_2V_2$

$0.1 \times 1 = N_2 \times 100$

$N_2 = 10^{-3}$

$\left[H^+\right] = \sqrt{ka.c} = \sqrt{10^{-7} \times 10^{-3}}$

$= 10^{-5}$

$pH = 5$

46. I belongs to strong electrolytes

II belongs to weak electrolytes

47.

	C	H	N
	9	1	3.5
	$\frac{9}{12}$	$\frac{1}{1}$	$\frac{3.5}{14}$
	$\frac{3}{4}$	$\frac{1}{1}$	$\frac{1}{4}$
4X)	3	4	1

EF C_3H_4N

$n = \frac{MW}{EFW} = \frac{108}{54} = 2$

$MF = n(EF)$

$= C_6H_8N_2$

48. $2gm \quad - \quad 1cc$

$2000gm \quad - \quad 1000cc(1\ell)$

wt of solution vol.of solution

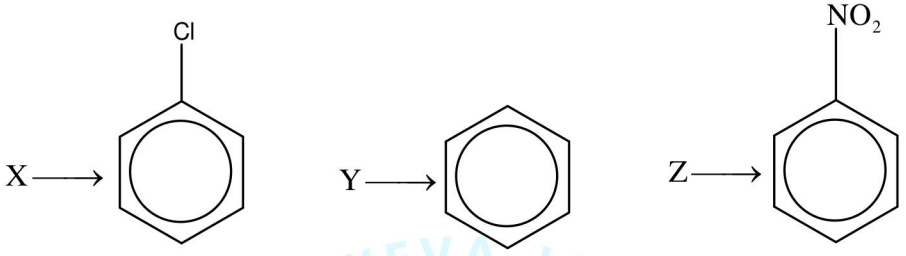
wt of solute = 1000 gm, wt of solvent = 1000 gm

$\therefore m = \frac{1000}{MW}\ \frac{1000}{1000} \qquad M = \frac{1000}{MW}\ \frac{1000}{1000}$

$\therefore \frac{m}{M} = \frac{1}{1}$

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49.



$MW\ C_6H_5NO_2 = 123$

50. $\frac{1}{2}N_2 + \frac{3}{2}H_2 \longrightarrow NH_3 \qquad \Delta H = -50kJ$

$-50 = (472.5 + 652.5) - 3x$

$x = 391.7 \cong 392kJ$

MATHEMATICS

51) Case I 5 _ _ _ 0 $N + 1 = 3^2.2^{12}$

$RNW = 39$

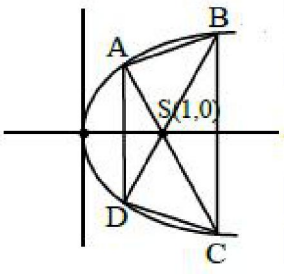
Case II 5 _ _ _ 1

5-----2
5-----3
6-----0
6-----1
6-----2
7-----0
7-----1

$N = 9 \times (8 \times 8 \times 8 \times 8) - 1$

52)

Sol.



$A\left(at_1^2, 2at_1\right) \& C\left(\frac{a}{t_1^2}, -\frac{2a}{t_1}\right),\ a = 2$

$Length\ AC = \left(t_1 + \frac{1}{t_1}\right)^2 = \frac{25}{4}, t_1 + \frac{1}{t_1} = \pm\frac{5}{2}$

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$\Rightarrow t_1 = 2 or \frac{1}{2}, A\left(\frac{1}{2},2\right), D\left(\frac{1}{2},-2\right), B(8,8), C(8,-8)$

So, area of trapezium $= 75$

53) There are 11 periods of in $(-12\pi, 10\pi)$ and in even period half of the time $\cos x$ is position hence probability is $\frac{1}{2}$

54) $f(k) = \frac{k}{2009}, f(2009-k) = \frac{2009-k}{2009}, f(k) + f(2009-k) = 1$

$$g(2009-k) = \frac{f^4(2009-k)}{(f(k))^4 + (1-f(k))^4} = \frac{(1-f(k))^4}{(f(k))^4 + (1-f(k))^4}$$

$g(k) + g(2009-k) = 1$

Hence $\sum_{K=0}^{2009} g(k) = \frac{2010}{2} = 1005$

55) $f(xy) = f(x).f(y) - f(x+y) + 1$

Put $y = 1 \quad f(x) = 2f(x) - f(x+1) + 1$

$f(x+1) - f(x) = 1$

$f(2) - f(1) = 1$ *Addision*

$f(3) - f(2) = 1$ $f(x+1) - f(1) = x$

$f(x+1) - f(x) = 1$ $\boxed{f(x+1) = x+2}$

$f(x) = x+1$

$1 + \sum_{K=1}^{\infty} \frac{f(K)}{2^K} = 4$

56) Let $P = (1,-1,0)$, $Q = (2,1,2)$ $R = (3,2,-1)$

Let $A(x,y,0)$

AP = AQ = AR

On solving $A = \left(\frac{3}{2},1,0\right)$

$A = \left(\frac{3}{2},1,0\right), B(1,3,-1), C = (2,0,2)$

AB = AC

Triangle ABC is an isosceles triangle

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Centroid $=\left(\frac{3}{2},\frac{4}{3},\frac{1}{3}\right)$

57) $R=\{(x,y)=x^2-4xy+3y^2=0, x,y\in N\}$

$x\in N: x^2-4x^2+3x^2=0 (x,x)\in R:$ Reflexive

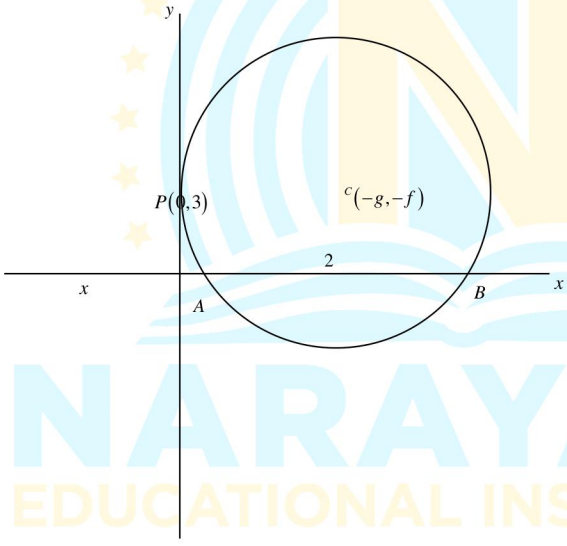
$(3,1)\to x^2-4xy+3y^2=0$

$(1,3)\to x^2-4xy+3y^2\neq 0$

'R' is not symmetric

R is not Transitive

58)



$x^2+2gx+(y-3)^2=0$

$x^2+y^2+2gx-6y+9=0$

$AB=2=2\sqrt{g^2-C}$ Centre $=\left(\sqrt{10},3\right)$

$1=g^2-C$

$1=g^2-9$

$g^2=10$

$g=\pm\sqrt{10}$

59) $T_n=\left(1+\frac{1}{2}+\frac{1}{2^2}+.....+\frac{1}{2^{n-1}}\right).\frac{1}{3^{n-1}}$

$=\left(1-\frac{\frac{1}{2^n}}{1-\frac{1}{2}}\right)\frac{1}{3^{n-1}}=\frac{2^n-1}{6^{n-1}}$

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$$Tn = 2\left(\frac{1}{3}\right)^{n-1} - \left(\frac{1}{6}\right)^{n-1}$$

$$S_{\infty} = \sum_{1}^{\infty} T_n = \frac{2}{1-\frac{1}{3}} - \frac{1}{1-\frac{1}{6}} = 3 - \frac{6}{5} = \frac{9}{5}$$

60) $f(x) = ax + b \Rightarrow f^1(x) = a > 0$

f is increasing in $[-1,1] \Rightarrow f(-1) = 0$

$f(1) = 2 \qquad \therefore f(x) = x + 1$

$\cot\left[\cot^{-1} 7 + \cot^{-1} 8 + \cot^{-1} 18\right]$

$= \cot\left[Tan^{-1}(1/7) + Tan^{-1}(1/8) + Tan^{-1}(1/18)\right]$

$= \cot\left[Tan^{-1}(1/3)\right] = 3 = f(2)$

61) $\frac{T_{r+1}}{T_r} = \frac{1}{3}, T_1 = a, T_2 = \frac{a}{3}, T_3 = \frac{a}{3^2}$

$T_7 = \frac{a}{3^6} = \frac{1}{243} \Rightarrow a = 3$

$T_r . T_{r+1} = \frac{a}{3^{r-1}} . \frac{a}{3^r} = \frac{a^2}{3^{2r-1}} = \frac{1}{3^{2r-3}}$

$\sum_{r=1}^{\infty} T_r . T_{rf+1} = 3 + \frac{1}{3} + \frac{1}{3^3} + \frac{1}{3^5} + \ldots\ldots =$

$= \frac{3}{1-\frac{1}{3^2}} = \frac{27}{8}$

62) Clearly, point R lies on the given line.

So, triangle RPQ is an isosceles triangle, i.e., RP = RQ

$RP = RQ = \sqrt{1+9+16} = \sqrt{26}$

Also, line is parallel to the vector $\hat{i} + \hat{j} - \hat{k}$,

$$\therefore \cos\theta = \frac{\left|\left(\hat{i} + 3\hat{j} - 4\hat{k}\right).\left(\hat{i} + \hat{j} - \hat{k}\right)\right|}{\left|\left(\hat{i} - 8\hat{j} + 2\hat{k}\right)\right|\left|\left(\hat{i} + \hat{j} - \hat{k}\right)\right|} = \frac{8}{\sqrt{78}} \Rightarrow \sin\theta = \sqrt{\frac{14}{78}}$$

$$\Rightarrow \sin 2\theta = 2\sin\theta\cos\theta = \frac{8\sqrt{14}}{39}$$

Therefore, area of triangle PQR = $\frac{1}{2}(RQ)(RP)\sin 2\theta = \frac{8\sqrt{14}}{3}$

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63) $f'(x)=\tan^2 x+k$ where $k=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} f(x)\,dx$

$f(x)=\tan x-x+Kx+C$

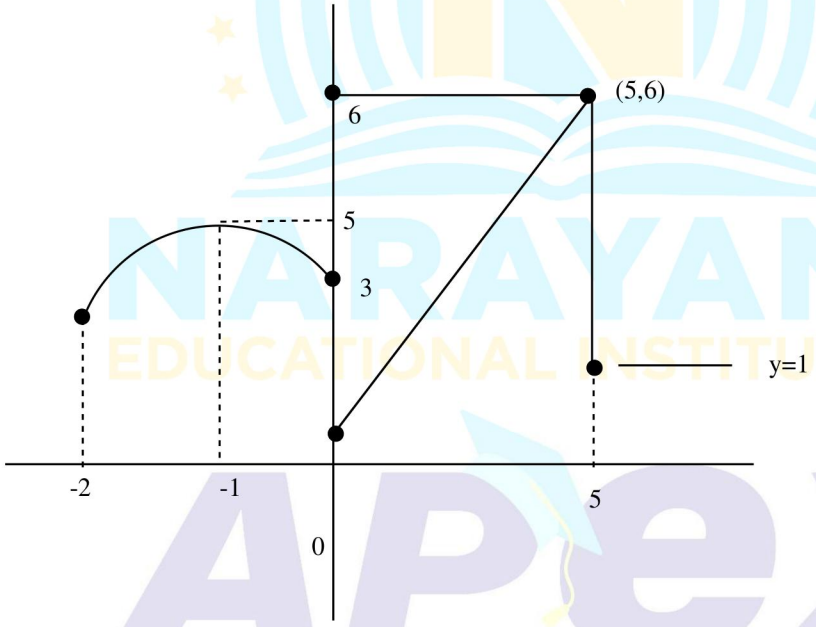
$f\left(\frac{\pi}{4}\right)=1-\frac{\pi}{4}+k\frac{\pi}{4}+C-\frac{\pi}{4}$

$\therefore f(x)=Tanx-x+kx-k\frac{\pi}{4}-1$

$k=\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} dx=-\frac{\pi}{2}-k\frac{\pi^2}{8}$

$\therefore K=\frac{-4\pi}{8+\pi^2}$

64)



65) $Case(1)$ $x\geq -2,\ x^2-2>0\Rightarrow |x|>\sqrt{2}$

Solution lies in $\left[-2,-\sqrt{2}\right)\cup\left(\sqrt{2},\infty\right)$

Case(2) $x\leq -2$

$x^2+2x+2>0$

$(x+1)^2+1>0$

Which is true for all real x, Hence solution set is $(-\infty,-2]$

$\because$ combing the two cases $x\in\left(-\infty,-\sqrt{2}\right)\cup\left(\sqrt{2},\infty\right)$

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66) $e^x F(x) = \int_3^x \left(3t^2 + 2t + 4F^1 t\right) dt$

Let $y = F(x)$

$\frac{dy}{dx} + \frac{e^x}{e^x - 4} y = \frac{3x^2 + 2x}{e^x - 4}$

$I.F = e^x - 4$

$Solution\ y\left(e^x - 4\right) = \left(3x^2 + 2x\right) dx$

$y\left(e^x - 4\right) = x^3 + x^2 + c$

$F(3) = 0 \Rightarrow y(3) = 0$

$c = -36 \quad F(x) = \frac{x^3 + x^2 - 36}{e^x - 4}$

67) Given, ${}^nC_{r-1} = 4, {}^nC_r = 12, {}^nC_{r+1} = 16$

$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{4}{12}$

$\frac{r}{n-r+1} = \frac{1}{3}$

$3r = n - r + 1$

$4r = n + 1 \ldots\ldots(1)$

$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{12}{16}$

$\frac{r+1}{n-r} = \frac{3}{4}$

$4r + 4 = 3n - 3r \ldots\ldots(2)$

Solving (1) and (2)

Then we get $n - \frac{23}{3}\ and\ r = \frac{7}{5}$

$C = \left(S\left(\frac{23-7}{5}\right)\right), S\left(\frac{23+7}{5}\right)$

$C = (16, 30)$

Centroid of ABC is α

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$$\alpha = \left(\frac{\cos t + \sin t + 16}{3}, \frac{\sin t - \cos t + 30}{3}\right)$$

$$\therefore (3x-16)^2 + (3y=30)^2$$

$$= \left[3\frac{ost + \sin t + 16}{3} - 16\right]^2 + \left[3\frac{(\sin t - \cot + 30)}{3} - 30\right]^2$$

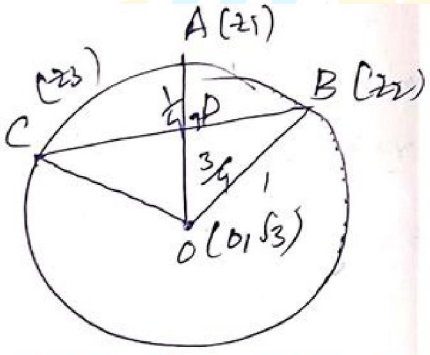
$$= [\cos t + \sin t + 16 - 16]^2 [\sin t + \cos t + 30 - 30]^2$$

$$= \cos^2 + \sin^2 t + 2\cos t \sin t + \sin^2 t + \cos^2 t - 2\cos t.\sin t$$

=

68)

:



$$|OA| = |OB| = |OC| = 1$$

$$\frac{3Z_1 + \sqrt{3}}{3+1} = \frac{Z_2 + Z_3}{2}$$

$$|OD| = \frac{3}{4}|AO| = \frac{3}{4}.1$$

ΔBDO

$$BD = \sqrt{1^2 - \left(\frac{3}{5}\right)^2} = \frac{\sqrt{7}}{4} \qquad AB = \frac{1}{4}|AO| = \frac{1}{4}.1$$

$\ln \Delta BAD$

$$|AB| = |z_1 - z_2| = \sqrt{\left(\frac{\sqrt{7}}{4}\right)^2 + \left(\frac{1}{4}\right)^2} = \sqrt{\frac{7}{16} + \frac{1}{16}} = \frac{1}{\sqrt{2}}$$

69) Let 'x' denote the number of kings in a draw of two cards

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X : O	1	2
$P(x)$: $\frac{188}{221}$	$\frac{32}{221}$	$\frac{1}{221}$

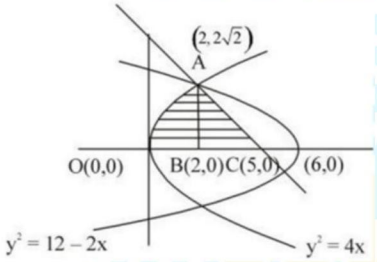
Mean of $X = \sum x_i P(x = x_i) = \frac{34}{221}$

$E(x^2) = \sum x_i^2 P(x = x_i) = \frac{36}{221}$

$Ver(x) = E(x^2) - (E(x))^2 = \frac{36}{221} - \left(\frac{34}{221}\right)^2$

$\sigma^2 = \frac{6800}{(221)^2}$

70) $x \geq 0, y \geq 0, y^2 \leq 4x, y^2 \leq 12 - 2x, 3y + \sqrt{8}x \leq 5\sqrt{8}$



Area $= \int_0^2 \sqrt{4x}\ dx + \frac{1}{2} \times 3 \times 2\sqrt{2}$

$= \left[2 \times \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^2 + 3\sqrt{2} = 2.\frac{2\sqrt{2} \times 2}{3} + 3\sqrt{2} \quad = 2.\frac{2\sqrt{2} \times 2}{3} + .\sqrt{2} = \frac{17\sqrt{2}}{3} \Rightarrow \alpha = \frac{17}{3}$

71) $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times 3}$

No. of elements in $S_1 = A = A^T \Rightarrow 7^6$

No. of elements in $S_2 : A = -A^T \Rightarrow 7^3$

No. of elements in $S_3 : a_{11} + a_{22} + a_{33} = 10$

$n(S_3) = 0$

$n(S_1 \cap S_2) = 1$

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$n(S_1 \cup S_2 \cup S_3) = 7^6 + 7^3 + 0 - 1 = 7^6 + 7^3 - 1$

Remainder is 6

72) $S = coeff.of\ x^4 in$

$$= \left[{}^4C_0 \cdot \left((1+x)^{101}\right)^4 - {}^4C_1 \left((1+x)^{101}\right)^3 + {}^4C_2 \left((1+x)^{101}\right)^2 - {}^4C_3 \left((1+x)^{101}\right)^1 + {}^4C_4 \right] - {}^4C_4$$

$$= \left((1+x)^{101} - 1\right)^4 - 1$$

$$= \left(1 + {}^{101}C_1 \times {}^{101}C_2 x^2 + {}^{101}C_3 x^3 + {}^{101}C_4 x^4 + \ldots\ldots - 1\right) 4 - 1$$

$$= x^4 \left({}^{101}C_1 + {}^{101}C_2 x + {}^{101}C_3 x^2 + \ldots\ldots\right)^4 - 1$$

$$= x^4 \left(101 + \left({}^{101}C_2 + {}^{101}C_3 x^2 + \ldots\ldots\right)\right)^4 - 1$$

$$= 101^4$$

$$k = 4$$

73) $\left|\bar{u}\right|^2 = \left|\bar{v}\right|^2 + 4\left|\bar{v} \times \hat{i}\right|^2 + 0 \qquad = v_1^2 + v_2^2 + v_3^2 + 4v_2^2 + 4v_5^2$

$= v_1^2 + 5\left(v_2^2 + v_3^2\right) \qquad \geq 2^2 + 5\left(1^2 + 0^2 = 9\right)$

$= \left|\bar{u}\right| \geq 3$

74)

$$f(x) = \begin{cases} 1+x \ if\ -1 \leq x \leq 0 \\ 1-x \ if\ 0 \leq x \leq 1 \\ 0 \ if\ x > 1 or\ x < -1 \end{cases} \qquad g(x) = f(x-1) + f(x+1) = \begin{cases} 0 & if\ x < -2 \\ 1 - |x+1| & if\ -2 \prec x \leq 0 \\ 1 - |x-1| & if\ 0 \leq x \leq 2 \\ 0 & if\ x > 2 \end{cases}$$

Points where f is not differentiable are -2, -1, 0,1 and 2.

75) $\left(\sqrt{a^2e^2 + b^2}\right)^2 \left(\sqrt{a^2e^2 + b^2}\right)^2 (2ae)^2$

$2.\left(a^2e^2 + b^2\right) = 4a^2e^2 \qquad \Rightarrow e^2 = \frac{b^2}{a^2} \ldots\ldots(1)$

$\therefore e^2 = \frac{1-b^2}{a^2} = 1 - e^2 \qquad \therefore e^2 = \frac{1-b^2}{a^2} = 1 - e^2$

$e = \frac{1}{\sqrt{2}} \Rightarrow \frac{\alpha}{\sqrt{\beta}} = e \qquad \therefore \text{Then } \alpha + \beta^2 = 1 + 2^2 = 5$

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